



Revealed social preferences and joint forest management outcomes[☆]



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ABSTRACT

We explored the role of revealed social preferences in the variation of JFM outcomes. To find out the social preferences of villagers, public goods game was organized in five randomly chosen villages in central India. At the village level, strong positive correlation was found between the average allocation to public good in the game and success of JFM in the village. In two villages, which had high success in JFM, villagers' social preferences were reciprocity, while in three other villages, which had moderate to low success in JFM, villagers' social preferences were commitment. We also found that demographic, economic, and social factors did not consistently explain individuals' preferences.

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1. Introduction

Joint forest management (JFM) was adopted in India to arrest the large scale forest degradation in state owned forests. The root cause of forest degradation can be traced back to about two centuries long history of exclusionary forest policies during the colonial rule. After India's independence in 1947, Indian government tried to redefine social objectives of forest management but primary objective continued to be the maximization of commercial timber yield. Until JFM was adopted in 1990, the management of forests was based on regulatory and authoritarian management practices, characterized by the exclusion of local community (Prasad and Kant, 2003).

JFM symbolises the idea of collaborative forest management (CFM) between local community and the state forest department, whereby the roles, responsibilities, decision making authorities, and benefits of both parties are spelt out (Bhattacharya et al., 2010). Although state-specific variations exist, many features of JFM institution are common. Under JFM, forest management unit is formed at village level, which is called by different names such as Joint Forest Management Committee (JFMC), Forest Protection Committee (FPC), and Eco Development Committee (EDC). These village committees (VCs) are required to perform planning, protection and other forest management operations. In India about 1.1 million VCs exist, which are managing and protecting about 22.9 million hectares of forests spread across 28 states and union territories (Lele, 2014).

The membership of committees varies across states but broadly each and every family, living near the forest under consideration, is eligible for membership. The participation in JFM is voluntary. One striking aspect of JFM is that although rights and duties have been given to VCs, the ownership of land remains with the forest department. A participating member of VC has free access to non-nationalized non-timber forest products (NTFPs) such as fuelwood, fodder, leaves, and fruits for personal use. Members also get some benefit from intermediate yield obtained from thinning and other silvicultural operations. Besides, VC gets a share from the final timber harvest, which is distributed equitably among the members. VC is collectively responsible for management and protection of forest from grazing, fire, theft or other damages and to report forest offenses to forest department.

Studies to evaluate the effectiveness of JFM indicate that the success of JFM varies across villages (Chakrabarti et al., 2005). The variation has been attributed to many factors including homogeneity of community (Kant, 2000; Kumar, 2002), community size (Behera, 2009), devolution of power to community (Singh et al., 2011), degree of dependence of community on forests (Kant 2000; Lise 2000), and condition of forest (Lise 2000; Kumar 2002; Behera 2009). Institutional factors such as structural resistance within the forest department (Kumar and Kant 2006; Lele 2014) and lack of recognition of the rights of local communities by the forest department (Sarker, 2009; Singh et al., 2011) have also been recognised as crucial for JFM success.

Some studies have concluded that social factors are more important than economic factors for the success of JFM (Lise, 2000), and presence of social capital enhances the success of JFM (Behera, 2009). Prasad and Kant (2003) have observed that individuals create non-market institutions to achieve socially optimal goal when they are faced with the realization of social gains. These studies allude to the importance of social preference in the success of JFM.

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It is pertinent to realise that JFM is characterised, among other things, by the interactions between community as a whole and the state, and more importantly interaction among the members of VC within a village community. Such interactions can be regarded as strategic because given the available resources every member wants to maximise his utility¹ from JFM. Game theory, which is used to model interactions between two or more players, can be useful in modeling the JFM and explaining the outcomes. However, use of game theory in modeling and explaining the outcomes of JFM has been scarce. Moreover, most of the work done so far is based on the classical game theory, i.e. it assumes that each individual is self-interested utility maximizer, e.g. (Kant and Nautiyal, 1994; Lise, 2001; Lise, 2005), though Shahi and Kant (2007, 2011) used evolutionary game theory to analyze the JFM, and Shahi and Kant (2013) used asymmetric public goods game to show that preferences of the forest user groups in three Indian states do not conform to rational economic agent.

Several empirical findings consistently challenge predictions that are based on the presumption of rational economic agent (*Homo economicus*), particularly when humans interact in collective action dilemma outside market settings. In recent years, many economists, psychologists, and sociologists have challenged the assumptions of rational choice theory (e.g. Colander, 2000; McFadden, 1999; Kant, 2005; Gowdy and Erickson, 2005; Gintis, 2000). Sen (1977) described rational economic agent as 'social moron' or 'rational fool', while McFadden (1999) declared that *Chicago Man*² is endangered species and economists are facing challenge to evolve *Chicago man* into *K-T Man*.³ A large number of laboratory and field experiments have already provided enough evidence to conclude that people also have social preferences (Cartwright, 2011). Ostrom (2005) opines that some people may be rational egoist, but some follow or can learn norms of such behaviour as reciprocity, fairness, and trustworthiness, particularly in the situations of social dilemmas. At the community level JFM presents such social dilemmas.

In this paper we formulate JFM as a linear public goods game and argue that though it is not optimal for a self-interested individual to participate in JFM, participation can be explained using theories of social preferences such as reciprocity, commitment, and altruism. We, therefore, hypothesise that people participate in JFM because they have social preferences, and the presence of social preferences can explain the success of JFM in a village. We use the data collected from five randomly selected villages in Seoni district of the state of Madhya Pradesh, India to validate our hypothesis. A survey questionnaire was used to collect demographic and socioeconomic data, and a linear public goods game was organized to get data on revealed social preferences. We analyse the data of public goods game using random effect model. We discuss the results of our analysis and address the validity concern of our study. The rest of the paper is organised accordingly.

2. JFM as a public goods game

JFM is implemented in those villages that have degraded forests in surrounding areas. The villagers have to forego extraction of forest resources and provide protection to the forest to help it regrow. For a household, the meaning of participation in JFM is to give up extraction of forest resources (e.g. timber, fuelwood, cattle grazing and NTFPs) unless allowed by the JFM bylaws, and to protect the forest from fires and illegal logging. Households receive benefits from the increased availability of fuelwood and fodder, and a share from intermediate felling and/or the final felling of timber. Both foregoing of unrestricted extraction of forest resources and protection activity can be considered as costs that

a household bears as result of participation in JFM. Households in the village are thus playing a public goods game—they are contributing to create forest biomass, which is expected to provide them benefit in future.⁴

If a household $i \in N$, where $N \equiv \{1, 2, \dots, n\}$ participates in JFM by choosing a participation level p_i : $0 \leq p_i \leq 1$, where '0' means no participation at all and '1' means full participation, utility of a household i from the participation in JFM can be written as:

$$u_i(p_1, p_2, \dots, p_n) = \frac{\theta_i}{n} \left(\sum_i p_i \right) - p_i. \quad (1)$$

Factor $\frac{\theta_i}{n}$ can be interpreted as marginal per capita return (MPCR) for household i from the participation in the JFM. MPCR may vary across households due to endogenous time preferences and variations in household's utility from consumption of forest resource. Utility of a household from the consumption of forest resource, and therefore MPCR, is expected to be high for highly forest-dependent households.⁵

2.1. Rational choice theory and JFM outcome

As long as $1 < \theta_i < n$, under the assumption of rational choice theory, it is not optimal for household i to choose a participation level other than zero, i.e. equilibrium participation $p_i^* = 0$, $\forall i \in N$, though for the community as whole, it is optimal that everyone chooses the highest participation level i.e. $p_i^* = 1$, $\forall i \in N$. The typical value⁶ of θ_i in the JFM villages is between 1 and n , and hence it is not optimal for a self-interested rational household to participate in JFM.

2.2. Theories of social preference and JFM outcome

A number of theories and associated models have been proposed to explain social preferences. Three theories of social preferences – i.e. altruism, commitment, and reciprocity – are more appropriate in JFM context than other⁷ theories. We use the framework developed by Croson (2007) for our analysis.

2.2.1. Commitment

As per commitment theory, an individual member in a group behaves in a manner as he would like others to behave, irrespective of the actual behaviour of others; and this behaviour of the individual is referred as "Kantian" (Laffont, 1975), "rational commitment" (Harsanyi, 1980), or "principle of unconditional commitment" (Sugden, 1984). Hence, as per commitment theory, a household will choose a level of participation as it desires every other household to choose, disregarding the actual participation levels of other households. In JFM game, household maximizes its utility given in Eq. (1) subject to the belief that $p_i = p_j \forall j \neq i$. Hence, individual's utility becomes:

$$u_i(p_1, p_2, \dots, p_n) = \theta_i p_i - p_i \quad (2)$$

For $\theta_i > 1$, it is optimal for the household i to choose positive equilibrium participation ($p_i^* > 0$). Since the participation level of a household does not depend on the actual participation level of other households, we get $\left(\frac{\partial p_i^*}{\partial p_j} = 0 \forall j \neq i \in N \right)$.

⁴ Since households have to incur the cost from the beginning and benefits begin to flow only at later date, the time preference of the villagers also become an important factor in their decision to participate in JFM. We are, however, not explicitly addressing the issue of time preference in this paper.

⁵ MPCR will also be high for those households, which are relatively patient i.e. have low discount rate.

⁶ We have calculated these values for the five villages from the socio-economic data collected during the study.

⁷ Theory of inequality aversion is one such theory of social preferences. Please refer to the Fehr and Schmidt (1999) model for the case with complete information, and the Bolton and Ockenfels (2000) model for case with incomplete information.

¹ Utility is based on preference of the individual, but preference might not be self-interested.

² Author describes a consumer *Chicago man* if he conforms to the standard economic model of perception, preference, and process rationality.

³ Since many of the features of the psychological foundations of decision making are rooted in the work of Kahneman and Tversky (e.g. Tversky and Kahneman 1974; Kahneman and Tversky 1979), the author calls this view the *K-T man*.

2.2.2. Altruism

Theories of altruism assume that individuals care directly about the consumption or utility of others, and therefore the utility of others appears positively as an argument in an individual's utility function (e.g. Becker, 1974). Andreoni (1990), in addition to welfare of others, included 'warm glow' of giving in utility function and called it impure altruism. In JFM game, under pure altruism household i maximizes its utility given by Eq. (3). The first argument of the utility function is household's own payoff and the second argument is payoff of the entire village.

$$u_i(p_1, p_2, \dots, p_n) = u_i\left(\left[\frac{\theta_i}{n} \left(\sum_i p_i\right) - p_i\right], \theta_i \sum_i p_i\right). \quad (3)$$

Assuming both personal consumption and altruistic consumption are normal goods with decreasing returns, whenever condition for public goods game is satisfied i.e. $1 < \theta_i < n$, the optimal participation level is positive for each household ($p_i^* > 0$) (Croson, 2007). However under this assumption, level of participation for household i is negatively correlated ($\frac{\partial p_i^*}{\partial p_j} < 0 \forall j \neq i \in N$) with that of other households (Sugden, 1982).

This is known as crowding out effect in pure altruistic theory.

Under impure altruism theory, households maximize utility function, which includes the above two arguments as well as a third argument – the level of participation (p_i) by the household itself. The utility function of the household can be written as below:

$$u_i(p_1, p_2, \dots, p_n) = u_i\left(\left[\frac{\theta_i}{n} \left(\sum_i p_i\right) - p_i\right], \theta_i \sum_i p_i, p_i\right). \quad (4)$$

The third argument of the utility function causes warm glow effect. Assuming that all three consumptions, i.e. personal consumption (1st argument), altruistic consumption (2nd argument) and warm glow consumption (3rd argument) are normal goods with decreasing returns, whenever conditions for public goods game are satisfied, i.e. $1 < \theta_i < n$, the optimal level of participation is always positive ($p_i^* > 0$) (Croson, 2007). Like pure altruism case, chosen participation level of household i is negatively correlated with that of other households ($\frac{\partial p_i^*}{\partial p_j} < 0 \forall j \neq i \in N$). In the case of impure altruism this is known as 'partial crowding out' effect, which is less than the pure altruism case (Andreoni, 1989).

2.2.3. Reciprocity

In reciprocity people respond kindly to the perceived kind actions, and hostilely for perceived hostile actions of others (Fehr and Fischbacher, 2005). Reciprocity is fundamentally different from altruism, which is a form of unconditional kindness (Fehr and Gächter, 2000b). Various models of reciprocity have been developed by many scholars (e.g. Rabin, 1993; Levine, 1998; Dufwenberg and Kirchsteiger, 2004; Falk and Fischbacher, 2006).

As per Sugden's (1984) model of reciprocity, which is based on moral obligations, an individual maximizes his/her utility subject to the constraint of the principle of reciprocity. According to this theory, the household maximizes its utility given by Eq. (1) subject to the condition that $p_i = \min(p_i^c, p_j \forall j \in N)$, where p_i^c is the same level of participation as household i would be choosing under the commitment theory. Whenever condition for public goods game is satisfied, i.e. $1 < \theta_i < n$, the optimal participation level is always positive ($p_i^* > 0$) (Croson, 2007). Under the assumption of reciprocity theory, the equilibrium participation level of household i increases with the increase in the participation level of other households of the village ($\frac{\partial p_i^*}{\partial p_j} > 0 \forall j \neq i \in N$) (Sugden, 1984).

Optimal participation levels and correlations between equilibrium participation levels of a household and participation levels of other households in a village for different social preferences are given in Table 1.

3. Methodology

The study involved the selection of villages, determination of social preferences of households and their social, economic, and demographic features, and the determination of the status of JFM in the selected villages. The methods used for each of these four steps are discussed next.

3.1. Selection of villages and their brief description

Site for the collection of data was selected in Seoni district of state of Madhya Pradesh in central India. The state is endowed with rich and diverse forests, and is one of the pioneer states in JFM. Forests are almost exclusively owned by the state. In the district, people living in and around forests are tribal and depend on forests for their subsistence needs and cash income. They had been living there for many generations and forests have become a part of their cultural identity. JFM committees have been formed in almost all villages near forest areas in the district. Two blocks of the district were selected randomly, and out of 203 villages in these two blocks, five villages were selected randomly for the study. The selected villages were Bharda, Dhundhi, Jwara, Pahari, and Rahlon.

Bharda and Dhundhi are exclusively inhabited by a scheduled tribe but there is no cohesion among community members in either village. In Bharda, most young men migrate to cities for labour while in Dhundhi, most people practice agriculture and only few young men migrate to cities. Forests in both villages are highly degraded. Illegal collection of fuelwood for cash income is the main reason of degradation in Bharda, while uncontrolled grazing is the driving force in Dhundhi. In Bharda, people collect fuelwood from forests of other villages, while in Dhundhi they use agriculture residue and trees from agricultural fields for fuelwood. JFM committees are inactive and villagers are hostile to

Table 1

Optimal participation level of a household and correlation with the participation levels of other households for different theories of the social preference.

Theory	Form of the utility function $u_i(p_1, p_2, \dots, p_n)$	Optimal participation level of a household (p_i^*)	Correlation with the participation level of other households
Self-interested rational	$\frac{\theta_i}{n} (\sum_i p_i) - p_i$	Zero	None
Commitment	$\frac{\theta_i}{n} (\sum_i p_i) - p_i$	+ ve	None
Pure altruism	$u_i\left(\left[\frac{\theta_i}{n} (\sum_i p_i) - p_i\right], \theta_i \sum_i p_i\right)$	+ ve	– ve
Impure altruism	$u_i\left(\left[\frac{\theta_i}{n} (\sum_i p_i) - p_i\right], \theta_i \sum_i p_i, p_i\right)$	+ ve	– ve
Reciprocity	$\frac{\theta_i}{n} (\sum_i p_i) - p_i$	+ ve	+ ve

Source: Adapted from Croson (2007)

forest department in both villages, though degree of hostility is slightly less in Dhundhi than that of Bharda.

Rahlon is a heterogeneous community and includes members of scheduled tribe, scheduled caste, and backward classes. Occupations of these social groups are not similar, but all live cohesively. JFM committee, which had been non-functional until four years ago, has been revived. Forest of the village has also improved in last four years. Uncontrolled grazing was the main reason of forest degradation, which had been stopped. People do not get many benefits from forest because it is still young, but they take lot of interest in JFM affairs.

Jwara and Pahari have majority of scheduled tribes with few households belonging to scheduled caste and backward classes. Despite the difference in castes, communities are cohesive. Most households practice subsistence agriculture and some young men migrate to cities for wage labour. In Jwara, forests condition is very good, people get fuelwood and timber from forest as per JFM resolution. JFM committee is well functioning and very good coordination exists between the villagers and forest officials. In Pahari, forest condition is moderate and villagers get limited amount for fuelwood from forest. Only some people take interest in the affairs of JFM. Coordination with the officials of the forest department is also moderate.

3.2. Collection of demographic and socioeconomic data

A questionnaire was used to collect household data. The questionnaire included questions on size, income, and caste of household; age, gender, and education level of the head of household; and participation, benefits received, and expectations from JFM. The questionnaire was administered to all willing households in five villages.

3.3. Evaluation of JFM success

Ever since the concept of sustainable forest management (SFM) gained importance in the late 1980s, a need was felt to develop criteria and indicators (C & I) to monitor the progress of SFM (Kotwal et al., 2008). There are nine major regional and international processes for developing C & I for measuring SFM (Castañeda, 2000). Hence, we also use C & I approach to measure the state of JFM.

The Bhopal-India process, which caters to the Indian forests, has 8 criteria and 43 indicators. However, the C & I developed for SFM cannot be applied directly to JFM due to the scale and more focus on peoples' participation in JFM. We modified C & I suggested by Chattopadhyay and Datta (2010) for the evaluation of functioning of forest protection committees (FPC) in West Bengal, India as per the conditions of our site. The set of 4 criteria and 12 indicators used in our study are given in Table 2. Five members of research team involved in data collection assessed the values of each indicator for each village and assigned the weights to each indicator. Each member then calculated the total score of the state of JFM for each village on the scale of 1–10. The overall

state of JFM for each village was calculated by taking average of the scores of five members.

3.4. Public goods game

We used a public goods game, designed as a JFM game, to find the social preferences of the heads⁸ of households. We used the design of the public goods game that has been used by Andreoni (1988), Fehr and Gächter (2000a), and Croson (2007).

In each village, all the willing heads of households were recruited to play the game. The subjects were randomly divided into groups of eight individuals. The rules of the game were explained⁹ verbally to each group. After explaining the rules, the group of eight individuals was randomly divided into two groups of four individuals each, and these groups were seated in separate enclosures. Each group of four individuals was endowed with 100 tokens, which was divided equally among four members. Thus each of the four members had 25 tokens. The tokens were given to the subjects in paper wraps. The subjects were told that the game was about their forest, and trees have been replaced by tokens only because trees could not be accommodated in the room. Four rounds of the game were played in Jwara village and five rounds in other four villages.¹⁰ In each round, two groups of four members were randomly formed within the group of eight individuals. Thus each subject was involved in four one shot games in Jwara village and five one shot games in other four villages.

In each round, subjects were asked to sit facing each other and they were specifically asked to note who their group members were. They were not allowed to talk or communicate in any manner until all rounds were completed. Subjects were asked to turn around and face the wall and pick as many tokens as they wish in their hands and leave the rest in the paper wrap. The subjects were instructed not to show their picked tokens to other members or try to see how many tokens were being picked by other members. The number of tokens picked by each member was recorded by the researchers but it was not told to the other group members. Subjects were then asked to place the picked tokens in the paper wrap, and turn around and sit facing each other.

The tokens picked up by the subjects went to their personal accounts and the tokens left in the paper wrap went to the group account. Subjects received ₹1 for each token in their personal accounts, and the group as a whole received ₹2 for each token in the group account, which were equally divided among the four group members. This yielded a marginal per capita return (MPCR) of 0.5, same as in some previous public goods game experiments (e.g. Andreoni, 1988; Croson, 2007).

3.5. Econometric model to determine social preferences of a village

JFM decision making is based on interactions among peoples of a village, and social norms with respect to JFM may vary from village to village. Hence, we estimated social preferences of every village, and not of the pooled population of five villages, but we pooled the data to take advantage of higher number of degrees of freedom. Social preferences were determined by regressing the contribution made by the individual i in round t of the game on the total contribution made by the other three members of the same group in the same round. To capture the variations among villages, village dummy variables were interacted

Table 2
Criteria and indicators for evaluation of JFM.

Criteria	Indicators
1. Peoples' participation	1.1 Involvement of people in the affairs of JFM committee 1.2 Influence of the chairperson of the JFM committee over the villagers 1.3 Mutual cooperation among the members of the village
2. Coordination with forest staff	2.1 Acceptance of forest staff in the village 2.2 Meetings of forest staff and the villagers 2.3 People's trust in the motive and action of forest staff
3. Condition of the forest	3.1 Success in stopping illegal activities in the forest 3.2 Recent improvement in the growing stock of the forest 3.3 Overall condition and health of the forest
4. People's attitude	4.1 Peoples' attitude towards the protection of forests 4.2 Peoples' awareness about the importance of forests 4.3 People's willingness to protect the forest for future

⁸ Generally, in JFM, decision making is done by household's head, and therefore only household's head were recruited for the public goods game.

⁹ Three examples were used to demonstrate the game to participants. In the first example very few tokens were picked, in the second almost half of tokens were picked, and in the third almost all tokens were picked. This was purposely done to avoid any anchoring bias. Subjects were also clearly explained that their total payoff will depend on the number of tokens picked by them as well as by the number of tokens left by their group members.

¹⁰ After data collection in Jwara, standard error of the estimation was calculated, and it was decided to do five rounds in other villages to keep the standard error small enough for meaningful inferences.

with the contribution made by the other three members of the group. Social preferences of an individual may vary with the social homogeneity of the groups as well as with some socioeconomic characteristics. We included social homogeneity and five socioeconomic variables¹¹ as control variables. A group was defined as a homogenous group when all four member of the group belong to the same social group i.e. scheduled tribe, scheduled caste, or backward class, and a dummy variable was used for social homogeneity. All these six variables were interacted with village dummy variables to capture variations across villages. Similarly to capture variation in different rounds, round's dummy variables were interacted with village's dummy variables. The model specifications are given in Eq. (7) and explanation of all variables is given in Table 3.

$$\begin{aligned} Cont_{it} = & \alpha_0 + \sum_{j=1}^{j=4} \alpha_j 1[Vill_j] + \sum_{j=1}^{j=5} \beta_j ContG_{it} \times 1[Vill_j] \\ & + \sum_{j=1}^{j=5} \gamma_j ContG_{it} \times 1[Vill_j] \times 1[HomSoc] \\ & + \sum_{k=1, j=1}^{k=5, j=5} \delta_{kj} SocEco_k \times 1[Vill_j] + \sum_{l=1, j=1}^{l=4, j=4} \eta_{lj} 1[round_l] \\ & \times 1[Vill_j] + u_i + e_{it}. \end{aligned} \quad (7)$$

As shown in Eq. (7), we included two error terms. The variable u_i captures all unobserved time-constant factors that affect contribution of individual i . The term e_{it} is an idiosyncratic error, which represents time varying factors that affect contribution of individual i in round t of the game. Random effects¹² regression method was used to estimate the model.

4. Results and discussion

4.1. Socioeconomic characteristics of the participants in public good game

A total of 336 heads of the household (out of 424 households) from the five villages participated in the game. The basic statistics of the socioeconomic and demographic features of the participants are given in Table 4.

Most subjects are males because generally head of household is the eldest male member except when eldest male member either migrates in search of work or dies. In all villages, except Rahlon, scheduled tribe is the dominant social group. Average age (around 40 years) of the subjects is similar in all villages and age distribution is slightly skewed towards left.

Subject's years of education vary between 0 and 15 years with average between 3 and 5 years. Distribution of education is highly skewed towards left in all villages. Percentage of illiterate subjects is 52 in Bharda, 30 in Dhundhi, 52 in Jwara, 50 in Pahari, and 42 in Rahlon. The average annual household income varies between ₹33.6 (in Pahari) and ₹43.7 (in Dhundhi) thousands and is also slightly left skewed. There is not much difference in average household sizes – varies from 5.2 to 6.0 people.

4.2. Allocations in the public goods game and JFM outcomes

The number of participants, average number of tokens picked by individuals, percent of individuals picking >20 tokens, percent of

Table 3

Definitions of the variables used in the model of social preferences.

S no.	Variable	Definition
1.	$Cont_{it}$	Contribution made by individual i in round t of the game.
2.	$1[Vill_j]$	Indicator variable, which is 1 when individual i belongs to village j , 0 otherwise. $Vill_j$ has five unique values i.e. Bharda, Jwara, Dhundhi, Pahari, and Rahlon.
3.	$ContG_{it}$	Total contribution made by other three members of the group to which individual i belongs in round t of the game.
4.	$1[HomSoc]$	Indicator variable, which is 1 when individual i belongs to a group that has all the members from the same social group, 0 otherwise.
5.	$SocEco_k$	Five socio economic variables. Age: age of the individual participating in the game. Household size: Number of people in the household. Education: Number of year subject has been in the school. Male: Dummy variable, 1 if subject is male. Income: Household income of the subject (₹ in thousands).
6.	$1[round_l]$	Indicator variable, which is 1 when individual i is playing in round l of the game, 0 otherwise.
5.	u_i	Random error corresponding to each individual, which is assumed to be independent of all explanatory variables.
6.	e_{it}	Unexplained error in the contribution made by the individual i in round t of the game.

individuals picking <5 tokens, and JFM score for each village are given in Table 5.

The average numbers of tokens picked, which varies between 12.7 for Jwara and 20.2 for Bharda, show that on average people are neither completely self-regarding nor completely other-regarding. Most individual's preferences lie between the two extremes. The average number of tokens picked by the individuals of a village and JFM score of the village have very strong negative correlation (correlation coefficient = -0.96 , p -value = 0.01). Similarly, the percentage of individuals picking >20 tokens in a village – indicating preference leaning towards self-regarding extreme – is also negatively correlated with JFM score of the village (correlation coefficient = -0.92 , p -value = 0.03). These negative correlations imply that more self-regarding preferences in the village lead to less success in JFM. JFM score is positively correlated with the percentage of individuals towards the other-regarding extreme – picking <5 tokens – although correlation is not very strong (correlation coefficient = 0.67 , p -value = 0.2). This indicates that if there are more people in the village with other-regarding preferences, success in JFM is more. These results suggest that people's preferences exhibited in the proxy public goods game and their true preferences in JFM are interrelated.

A strong correlation between success in JFM and presence of other-regarding preference in the village should not be surprising. Although success of the JFM is attributed to many social factors, social preferences can be considered as synthesized expression of such variety of social factors. Social preferences are context specific and vary across individuals because everyone's synthesis of these factors is different. The average level of social preferences, however, in one social group (in our case village) is expected to be different from the other social group due to the preponderance of similar social factors.

4.3. Estimation result of econometric model

The coefficients of the estimated model and their clustered¹³ standard errors are given in Table 6.

¹¹ In other similar studies (Carpenter and Seki, 2011; Croson, 2007), authors did not include socioeconomic characteristics of the subjects in their models.

¹² Please refer to Wooldridge (2009, p. 489) for details on random effect regression. Random effect regression assumes that unexplained time-constant effect on individual contribution is uncorrelated with all the explanatory variables. In our model, since we have controlled fixed effects arising out of socioeconomic factors, we have reason to believe that error term u_i is uncorrelated with all the explanatory variables. Pooled OLS was another option, but Breusch and Pagan Lagrangian multiplier test for random effect rejected the null hypothesis $H_0: Var(u) = 0$, i.e. there is no panel effect ($\chi^2(1) = 1771.5, p = 0.000$). Finally, we used Hausman test to decide between random effect and fixed effects. We failed to reject null hypothesis H_0 : Difference in coefficient not systematic i.e. random effect is the preferred model ($\chi^2(27) = 29.57, p = 0.334$).

¹³ In our game we have randomly selected groups of eight individuals and then randomly divided the individuals in two groups of four in each round of the game. Since playing groups were randomised only within the groups of eight, we used the clustered standard error on 42 groups of eight. See Angrist & Pischke (2009, p. 319) for a discussion on clustered standard error.

Table 4
Summary statistics of demographic and socioeconomic data of surveyed people.

Village	Gender and social groups					Age (years)		Household income (₹000)		Education of head of household (years)		Household size	
	N	M	ST	SC	BC	Mean	Sd	Mean	Sd	Mean	Sd	Mean	Sd
Bharda	56	40	56	0	0	37.9	12.8	41.6	21.2	3.8	4.4	5.2	1.9
Dhundhi	72	68	72	0	0	43.0	14.9	43.7	29.8	4.9	3.8	5.8	2.0
Jwara	64	49	58	4	2	39.5	11.2	43.0	25.1	3.3	3.8	5.3	2.4
Pahari	72	68	63	0	9	40.0	12.1	33.6	26.0	3.6	4.1	5.7	2.1
Rahlon	72	60	28	41	3	42.3	11.4	35.3	16.0	4.1	4.1	6.0	2.6

N: Number of people surveyed, M: number of males, ST: scheduled tribe, SC: scheduled caste, and BC: backward class.

The model has positive intercept¹⁴ for all villages, which indicates that subject's preferences on an average are not completely self-regarding in any of the five villages. There is no systematic variation in the intercept across the rounds of the game in any village except for a minor change in round 2 in Bharda and in round 3 in Rahlon. This implies that the average level of contributions in each village remained the same in all rounds of the game. The correlation between individual contributions with the contribution of other three members of the group (denoted by β_3) indicates the type of preference that subjects on an average have in the villages. The coefficient is positive for Jwara and Rahlon and insignificant for Bharda, Dhundhi, and Pahari. Positive coefficient implies reciprocity and insignificant, i.e. zero, coefficient implies commitment. Incidentally Jwara and Rahlon also have the highest and second highest JFM scores and also the highest and second highest contribution in public goods game. This indicates that subjects in villages with successful JFM show higher other regarding preferences and reciprocity. In three other villages with low or moderate JFM score, subjects show lower other-regarding preferences and no reciprocity. High JFM scores, high other-regarding preferences, and reciprocity support the evolutionary psychologists' observation that human beings have the capacity to learn reciprocity norms and other social norms that increases returns from collective action (Ostrom, 2003). Lack of reciprocity in other three villages may be due to people's moderate involvement in JFM, which might have resulted in weaker reciprocity norms with respect to JFM that got reflected in their behaviour in the game.

The coefficient of social homogeneity¹⁵ of the groups is not significant for any village. This result confirms researchers' observation of no perceptible difference among the members of different social groups with respect to JFM.

Men contributed significantly higher than women in Bharda and Pahari villages. These results indicate that women in these two villages acted in more self-regarding manner than men. Experimental economics literature does not provide conclusive evidence of gender differences in other regarding preferences (Croson and Gneezy, 2009). However, in forestry literature, it is argued that women tend to manage forest resources sustainably because usually they are responsible for the collection of fuelwood and NTFP, and therefore they should have more representation in forest decision making (Agarwal, 2009b; Das, 2011). In contrast to this argument, Agarwal (2009a) found that women from landless households favoured for lenient rule in forest resource extraction. This shows that in the crunch even women behave in a self-regarding and impatient manner. It is pertinent to mention that almost all household in Bharda engage in illegal collection of fuelwood for cash

income. Farming is marginal and there are virtually no other sources of employment.

The coefficient of household size is negative and significant for Jwara and Rahlon villages. These results may indicate that the heads of the bigger households may demonstrate more self-regarding preferences due to higher forest resource requirements.

The coefficient of age is negative and significant for Dhundhi and Pahari villages. This result, for these two villages, supports young people's view stated during survey that forests are environmental resources and need to be conserved.

The significant negative coefficient of household income and education, in Dhundhi village, indicate that self-regarding preferences increase with income and education. A possible explanation of this relationship may be that people with higher income and education have more exposure and familiarity with market environment that is driven by self-regarding preferences (Bowles, 1998, p. 88); and there is strong complementarity between peoples' experiences in the real life and their preferences in the game (Gächter, 2007, p. 29). The absence of such correlation in other villages, particularly with education, could be due to the fact that most of the subjects have little¹⁶ or no education, and hence there is hardly any variation in the education level to yield a significant coefficient.

Estimation results also clearly point out that not all socioeconomic variables are significant for all villages. It is pertinent to mention that in some other cross cultural studies¹⁷ no consistent relationship between individual-level economic and demographic factors and subjects' behaviour in the games has been observed. In our model we have controlled socioeconomic variables separately for each village. Estimation result of the model and evidence from the previous studies validate our model. Subjects' preferences in the game are not expected to be explained by socioeconomic factors in all the villages uniformly and consistent manner, therefore it is necessary to include village-specific interactions in the model.

4.4. Are preferences in public goods game true reflection of preferences in JFM?

We found strong correlation between subjects' average allocations in the public goods game and JFM scores, high JFM scores in villages with reciprocity preferences, and low JFM scores in villages with commitment preferences.

It can be argued that people's preferences in the game may not be a true reflection of their preferences in real life. However, it is well established in experimental economics literature that experimental subjects use the moral sensibility and social beliefs that are shaped by their personal experiences in the real world in playing economic

¹⁴ α_0 corresponds to Dhundhi, ($\alpha_0 + \alpha_1$) corresponds to Bharda, ($\alpha_0 + \alpha_2$) corresponds to Jwara, ($\alpha_0 + \alpha_3$) corresponds to Pahari, and ($\alpha_0 + \alpha_4$) corresponds to Rahlon.

¹⁵ Out of total 404 groups that were randomly formed, 264 groups were found to be socially homogeneous. Since Bharda and Dhundhi have only one social group in the village, all groups in these villages are socially homogeneous. Because of multicollinearity the variable was dropped for these two villages. In Jwara, Pahari, and Rahlon 66%, 61%, and 8% groups were found to be socially homogeneous respectively.

¹⁶ About 75% of the subjects in other four villages received less than five years of education.

¹⁷ Henrich et al. (2005) conducted cross cultural studies in 15 small scale societies and found no consistent relationship between demographic and socioeconomic attributes of the subjects and their preferences in the games.

Table 5

Average number of tokens picked in public good game and JFM score of each village.

Name of the village	Number of participants	Average number of tokens picked	Percentage of individuals picking >20 tokens	Percentage of individuals picking <5 tokens	JFM score
Jwara	64	12.7*	11.3	16.0	7.7
Rahlon	72	13.3*	29.4	30.3	7.2
Pahari	72	15.6	28.1	7.5	5.6
Dhundhi	72	16.1	38.1	14.7	3.4
Bharda	56	20.2	64.6	3.6	1.7

* Two sample Wilcoxon rank-sum (Mann–Whitney) non-parametric test failed to reject the hypothesis at the 95% confidence level that there is no difference between token picked by the subjects in Jwara and Rahlon. For all other villages the test was conducted pairwise and hypothesis that there is no difference in average number of tokens picked in the two villages was rejected at the 95% confidence level.

games (Bowels and Gintis, 2011, p. 43). Deegen (2013) argued that people's choices could be a manifestation of an interaction between institutional structure and social order. In our study formal institutional structure to which subjects belonged was invariant, i.e. JFM guidelines are similar in all the five villages and for everyone. Institutional structure of the game and JFM did differ, but since subjects interacted in an extended social order, i.e. knew each other very well, it is very likely that they carried their beliefs from the JFM institution to the game institution.

Many studies, from diverse cultures, e.g. Lamelera whale hunters in Indonesia, the Orma herders in Kenya, and the Ache hunters in Paraguay, have shown that peoples' behaviour in the games and real life is correlated (Henrich et al., 2005). Glaeser et al. (2000) found that the subject who trusted the others in trust game also acted in a trusting manner in real life. Similarly, Karlan (2005) using trust and public goods games found that subjects, who acted as trustworthy in the experiment, were less likely to default in the repayment of microfinance loan.

Table 6

Estimated coefficients of the model of social preferences.

Dependent variable: individual contribution			
Variable	Coefficient	Variable	Coefficient
<i>Explanatory variables (clustered standard error shown within parentheses)</i>			
ContG · Bharda (β_1)	−0.004 (0.023)	Intercept (α_0)	18.287*** (2.688)
ContG · Dhundhi (β_2)	0.012 (0.021)	Intercept · Bharda (α_1)	−11.769* (4.629)
ContG · Jwara (β_3)	0.086** (0.027)	Intercept · Jwara (α_2)	−11.538* (4.582)
ContG · Pahari (β_4)	−0.048 (0.041)	Intercept · Pahari (α_3) §	−8.321 (4.361)
ContG · Rahlon (β_5)	0.036# (0.018)	Intercept · Rahlon (α_4)	−7.831 (6.464)
ContG · Jwara · HomSoc (γ_3)	0.009 (0.019)	Intercept · Bharda · R_2 (η_{21})	−0.633* (0.283)
ContG · Pahari · HomSoc (γ_4)	−0.004 (0.012)	Intercept · Rahlon · R_3 (η_{35})	1.064* (0.43)
ContG · Rahlon · HomSoc (γ_5)	0.02 (0.017)		
Male · Bharda (δ_{41})	4.356* (1.774)	hhSize · Jwara (δ_{23})	−0.44* (0.175)
Male · Pahari (δ_{44})	6.229** (2.087)	hhSize · Rahlon (δ_{25})	−0.75*** (0.187)
Income · Dhundhi (δ_{52})	−0.07** (0.025)	Age · Pahari (δ_{14})	−0.113* (0.058)
Education · Dhundhi (δ_{32})	−0.446* (0.191)	Age · Dhundhi (δ_{12})	−0.122** (0.047)

N = 1616, $R^2 = 0.2494$.Variables with insignificant ($p > 0.05$) coefficients are not shown in the table.* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.# $p = 0.053$.§ $p = 0.056$.

In the experiments involving common pool resources, Carpenter and Seki (2011) found that experimental measures of social preference in the Japanese fishermen can be directly linked to fishing productivity, and Fehr and Leibbrandt (2011) found that among Brazilian fishermen those who exhibited more cooperativeness and less impatient behaviour in the experiment were less likely to exploit the shrimps unsustainably. Similarly, in an experiment involving forest commons, Rustagi et al. (2010) found that among Bale Oromo people in Ethiopia the most common behavioural type is conditional cooperators (reciprocators) and the group with more number of conditional cooperators is more likely to be successful in managing the forest commons.

Though the field studies mentioned above lend support to the findings in our case study, the validity concerns of the experimental economics cannot be dismissed. We, therefore, discuss our results in the light of some of these concerns.

4.4.1. Validity concerns of the preferences derived through the game

The foremost concern of the experimental validity of games, as we have mentioned earlier also, is that games do not represent the real life. In our study we tried to make the games as close to real life as possible by telling the subjects that the game is about their forests and that they should treat the tokens as trees. Also, by asking the subjects to have a good look at their group members in each round and not revealing the choices made the subjects during the game, we tried to evoke individuals' belief about other members of the group. Unlike most of the other public goods experiments that have been conducted with university students and anonymously, our experiment was conducted with the subjects who all knew each other very well. Also in most of the other experiments people's choices were revealed to the subject at the end of each round. Such disclosure usually stimulates the subjects to react to the situation of the game. Our intentions, however, was not to find out how subjects react to other subjects' behaviour in the game but to find out their preference in JFM given their belief about other

subjects. A majority of the subjects picking different number of tokens in different rounds of the game is an indication that subjects invoked their held beliefs about other members during the game. Thus we believe that people preferences in the game, at least to some extent, were a reflection of their preferences in JFM.

The second concern regarding our result is that stakes in the game were not as high as in JFM. Binmore and Shaked (2010) have argued that the allocation in most games with money payoffs would be Nash Equilibrium, provided that the payoffs are sufficiently large and subjects have ample time for trial and error learning. In our game subject could get a payoff equivalent to about twice the prevailing daily wage rate. Although stakes may not seem very high, yet given the fact that villagers in question hardly have 15–30 days of employment in a year, stakes were sufficiently large to induce the subjects to play Nash Equilibrium strategy. Despite getting four to five opportunities of learning the game, subjects showed no inclination to move towards the Nash Equilibrium, i.e. picking all the tokens. The average number of tokens picked remained the same¹⁸ in all rounds (see Fig. 1). In most of the previous public goods experiments, it was found that participants tend to free ride in the last round. These experiments were conducted in anonymous settings and numbers of tokens picked by group members were revealed to the subjects after each round. In such experiments, subjects did not invoke their held beliefs about the other subjects but just reacted to the other subjects' behaviour in the game. In our game subjects could not react to the behaviour of other subjects because the number of tokens picked by the subjects was not revealed during the game. In addition, although the average number of tokens picked by individuals in a village remained almost the same in all rounds, tokens picked by a particular individual changed with the change in group members in different rounds of the game, which indicates that subjects did invoke their held belief about other subjects.

The third concern is that other-regarding preferences may be due to the subjects not understanding the game, and they cooperated just by mistake. This may be true for some subjects. However, in our experiment a strong correlation between the average number of tokens picked and JFM score indicates that majority of the subject did not make a mistake, rather they understood the game and their belief about the other members were by and large correct.

The fourth concern is that non-repeated face-to-face interactions that occur in a game are not part of everyday experience of the subjects, so they confuse experimental environment with repeated interactions of normal life and try to behave in such a manner as to build reputation for possible gains in future encounters. Even if it were true in our study, we will take it because the purpose of the experiment was to elicit the subject's preferences in JFM context. From the preferences of the subjects in the game, it appears that subjects did relate themselves to real life JFM game.

The fifth and a very different concern about the experimental evidence is the presence of framing effect, which are such factors that do not affect the payoff but do alter the preferences of the subject. As Andreoni (1995) points out, that when a public goods game is framed as an individual investment game in which subjects could invest in their private accounts, even at a cost to the other players, subjects show little cooperation. Such evidence from a number of other field experiments claiming that the experimenters can indeed manipulate the preferences of the subjects supports our view that the subjects had acted in the game as they would have acted in real life situation as the game was contextualised so. An individual conceptualizes the norms of social behaviour during the experiment and such norms are learned in social milieu (Ostrom, 2003). If the conceptualisation of norms in an individual can be manipulated to evoke the norm about the JFM

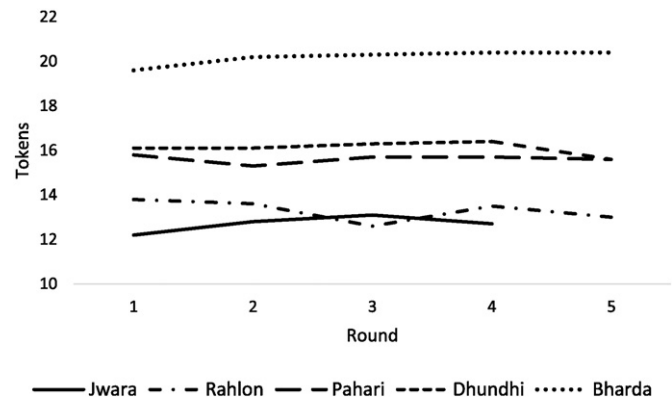


Fig. 1. Average number of tokens picked in each round.

during the public goods game, it is possible to elicit true behaviour of the individual.

The final concern is that the other-regarding behaviour in the game may have resulted from a desire to maintain reputation in the eyes of others. Rege and Telle (2004) showed that cooperation in a public goods game increases significantly when subjects were asked to write their contribution openly on a blackboard. In our study, it is very unlikely that subjects have taken action to build reputation in the eyes of other subjects, because their actions were kept secret and they played game with new group members in each round. One may still argue, though, that subjects behaved in a prosocial manner to maintain their reputation in researcher's eye. If subjects cared about the maintenance of reputation, it was their inner desire to be held in high esteem in the eyes of others. In our game, they knew that researchers would know what they did in the game. If the subjects were actually trying to boost their self-esteem by looking good in the eyes of someone they regarded as outsider, our point that subjects were having the moral preferences is reinforced. The variation in the moral preferences across the villages further corroborates that such desire to maintain one's reputation depends on the existing norms of the immediate social circle. Our previous discussion on the conceptualisation of norms and bringing in social experiences into the game implies that even if this concern is valid, it may have helped achieve the objective of the study.

5. Conclusions

We found that villagers' preferences lie between the two extremes of completely self-regarding and completely other-regarding. Our finding lends support to the growing criticism of the rational choice theory. Other key findings of our study are: (i) villagers' other-regarding preferences have strong positive correlation with the success of JFM in the village; (ii) villages with high other-regarding preferences and reciprocity have successful JFM; (iii) villages with moderate and low other-regarding preferences do not have reciprocity; and (iv) demographic variables do not consistently explain the peoples' other regarding preferences in all the villages.

Our study fills important gaps in the literature and validates existing literature. First, we have shown that variability in the outcomes of JFM can be explained through the variation of social preferences across villages. Some other studies have concluded that social factors explain the diverse outcomes of JFM. Most of the social factors mentioned in these studies affect the social norms of individuals and social preferences of individuals are combined manifestation of evolved norms in the social group to which they belong. Second, we use game theoretical analysis and experimental economics to explain JFM outcomes. We have gone beyond the framework of rational choice theory and observed the peoples' social preferences to explain the JFM outcome. The study adds to the existing literature on connection between real-life preferences and preferences in experiments. Third, the study also

¹⁸ Two sample Wilcoxon rank-sum (Mann–Whitney) non-parametric test failed to reject the hypothesis at the 95% confidence level that there is no difference in token picked by the subjects in any two rounds of a village.

validates the existing literature that individual-specific demographic and socioeconomic factors do not explain the other-regarding preferences in the game consistently.

The study, however, has certain limitations owing to time and logistic constraints. First, a larger sample size could have been more instructive. Second, >5 rounds of the game could have given enough data points to find individual's social preferences. Third, the playing groups' randomisation within a larger group, or even the whole village would have given better measure of correctness of beliefs and preferences.

There is scope to extend this research. First, the study can be replicated in varied settings so that results could be generalised. Second, game theoretical model of JFM should be developed and its equilibria should be analysed under the assumption of social preferences. Third, since prediction of the JFM outcomes in future is more important than explaining present condition of JFM, evolutionary game theory or agent based modeling approach should be used to predict the outcomes in future.

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